

意味役割誤分析仮説：第二言語英語における意味役割の誤付与と非
対格動詞の過剰受動化

**The Theta Misanalysis Hypothesis: Non-Target Theta Assignment and
Passive Unaccusatives in L2 English Acquisition**

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Abstract

The distinction between two classes of intransitive verbs, unaccusatives (e.g., *fall*, *appear*) and unergatives (e.g., *run*, *cry*), has been argued to exist in L2 interlanguage grammar. This claim is supported by evidence that L2 English learners frequently produce and accept unaccusatives in passive constructions, suggesting an underlying association between NP-movement in unaccusative and passive derivations. Specifically, learners appear to misanalyze NP-movement in unaccusative structures as requiring passive morphosyntax (*be* + *en*). While this analysis accounts for a range of learner behaviors, it fails to capture two crucial factors: discourse-pragmatic influences and the role of subject animacy (agentivity). To address these limitations, this study proposes the Theta Misanalysis Hypothesis—the classification of the sole argument in intransitive constructions (as either an internal or an external argument) is determined by the distribution of Proto-Role properties. In L2 grammar, discourse pragmatics and subject animacy contribute to Proto-Role assignments, influencing argument interpretation. To test this hypothesis empirically, an acceptability judgment task was conducted. The experimental results revealed no significant effect of context but showed robust effects of the lexical category. The findings are consistent with the Theta Misanalysis Hypothesis and offer new insights into the interaction between lexical semantics and syntax during L2 acquisition.

要旨

自動詞には、非能格動詞と非対格動詞の 2 種類が存在するとされており、この区別は第二言語の中間文法においても存在すると主張されている。この主張は、L2 英語学習者が非対格動詞を受動構文 (*be* + 過去分詞) で頻繁に産出・受容するという証拠によって支持されている。これは、非対格構文と受動構文における名詞句の移動が類似していることに起因し、学習者が両者を同一視している可能性を示唆する。すなわち、学習者は非対格構文における NP 移動を、受動形態 (*be* + *en*) を伴う構文として誤って解析している可能性がある。このような分析は、学習者の多様な構文処理を説明し得るものの、談話語用的要因や主語の生物性（動作主性）といった重要な要素を十分に捉えていない。こうした限界を補うため、本研究では意味役割誤解析仮説 (Theta Misanalysis Hypothesis) を提案する。この仮説では、非他動詞構文における唯一の項が内項か外項かを分類する際、プロト役割 (Proto-Role) 特性の分布が決定的な役割を果たすとされる。L2 文法においては、談話語

用的情報および主語の生物性がプロト役割の付与に影響し、それが項解釈に反映され则认为られる。この仮説の予測を検証するために、容認性判断課題を用いた実験を実施した。その結果、文脈的要因に関しては有意な効果は見られなかったが、語彙においては顕著な効果が確認された。これらの結果は意味役割誤解析仮説の予測と整合し、第二言語習得における語彙意味論と統語構造の相互作用に関する新たな知見を提供するものである。

1. Introduction

Single-argument verbs, traditionally considered intransitive, are now widely classified into two categories: unaccusative verbs (e.g., *fall*, *appear*) and unergative verbs (e.g., *run*, *cry*) (Burzio, 1986; Perlmutter, 1978). Although both classes share the canonical NP-V surface order in English, they differ in argument structure: the single argument of unaccusatives originates inside the VP as an internal argument typically bearing a Theme/Patient role, whereas that of unergatives is generated outside the VP as an external argument typically bearing an Agent role.

The phenomenon of split intransitivity has long captured the attention of researchers in first-language (L1) and second-language (L2) acquisition (e.g., Balcom, 1997; Hirakawa, 2001; Oshita, 2004; Randall et al., 2004; Shimada & Sano, 2007; Wang et al., 2024; Yuan, 1999). Passivization errors have been a long-standing issue in L2 acquisition (e.g., Hirakawa, 2000, 2006; Ju, 2000; Kim et al., 2025; Oshita, 1997, 2000; Zobl, 1986). As these errors predominantly arise with unaccusatives but rarely with unergatives, they are typically considered an indication that L2 learners are sensitive to the unaccusative–unergative distinction. Although many studies have focused on non-target passivization, a consensus remains elusive regarding the factors contributing to these errors.

Three major accounts have been proposed to explain passive errors in L2 grammars: discourse-pragmatic contexts (Ju, 2000), subject animacy (Chung, 2014), and the overgeneralization of passive morphosyntax (Oshita, 2000). Although these accounts aim to explain passive unaccusatives from three distinct perspectives—discourse-pragmatics, semantics, and syntax—their underlying assumptions differ. The first two analyses assume that the passivization of unaccusatives constitutes a form of verbal passivization, whereas the third assumes that it reflects the overt marking of NP-movement. This study critically evaluates these assumptions and argues that passive unaccusatives do not involve semantic passivization but rather reflect the use of passive morphosyntax to mark object-to-subject NP movement. A critical question arises regarding how the effects of discourse-pragmatic contexts and subject animacy can be accounted for without assuming the lexical causativization of unaccusatives.

To address this question, this study proposes *the Theta Misanalysis Hypothesis*, which integrates insights from the three existing accounts while overcoming their respective limitations. Based on the premise that passive unaccusatives reflect structural passivization without contributing to sentence meaning, the Theta Misanalysis Hypothesis posits that theta role assignment in L2 interlanguage deviates from that of the target grammar. Whereas the verb determines theta roles in the target grammar, the hypothesis suggests that theta roles in interlanguage grammar are influenced by contextual factors, the semantic properties of subjects, and the meaning of the verb. This perspective allows us to elucidate the effects of the discourse-pragmatic context and subject animacy on theta assignments.

The Theta Misanalysis Hypothesis generates two main predictions: (i) passive unergatives should occur when unergative verbs exhibit minimal agent-like properties; and (ii) discourse-pragmatic contexts should influence the acceptability of passive intransitives. An acceptability judgment task was conducted to test these predictions. The results supported prediction (i): passive unergatives belonging to non-controlled processes were rated higher than passive unergatives belonging to controlled processes, although they were rated lower than passive unaccusatives. However, prediction (ii) was not supported; learners' judgments of passive intransitives showed no sensitivity to contextual effects. Despite the absence of contextual effects, these findings are consistent with the Theta Misanalysis Hypothesis.

The remainder of this paper is organized as follows. Section 2 introduces the Unaccusative Hypothesis and discusses the syntactic and semantic differences between unaccusatives and unergatives. Section 3 reviews the Unaccusative Trap Hypothesis (UTH), which addresses the developmental path of L2 acquisition of unaccusatives, explores the longstanding debate surrounding the passivization of unaccusatives, and evaluates three competing accounts. This section also identifies the unresolved challenges faced in these analyses. Subsequently, Section 4 presents a novel hypothesis, referred to as the Theta Misanalysis Hypothesis, which resolves these problems, and discusses the plausibility of passive unergatives, challenging prior claims that unergatives categorically resist passivization. Section 5 reports the findings of the acceptability judgment experiment designed to test the hypothesis, and Section 6 discusses the implications of

the proposed hypothesis and directions for future research. Finally, Section 7 concludes the paper.

2. Unaccusative Hypothesis

According to the Unaccusative Hypothesis (Burzio, 1986; Perlmutter, 1978), intransitive verbs fall into two distinct classes: unaccusatives and unergatives. On the surface, their sentence structures appear similar in English, as shown in (1).

- (1) a. Miki walked. (unergative)
b. Miki arrived. (unaccusative)

Although both sentences in (1) exhibit the canonical NP-V word order, they differ significantly in their underlying syntactic structures. In (1a), the subject is an external argument base-generated outside the VP, as illustrated in (2a). Conversely, the subject in (1b) originates as an internal argument within the VP and subsequently moves to the subject position, as shown in (2b).

- (2) a. Miki [_{VP} walked]
b. Miki_i [_{VP} arrived *t_i*]

In unergative constructions, the subject is base-generated in the subject position. In unaccusative constructions, the subject originates as the verb's complement and moves to the surface subject position via NP-movement, leaving behind a trace/copy. In languages such as English, unaccusative constructions resemble passive constructions in that both involve NP-movement. However, a key distinction lies in the presence of a suppressed agent in passives, which unaccusatives lack. Moreover, NP-movement in passive clauses is morphologically marked by *be* + *-en*, while no such overt morphology is present in unaccusatives.

The unaccusative–unergative contrast is claimed to be semantically determined but syntactically encoded (Levin & Rappaport Hovav, 1995; Perlmutter, 1978; Sorace,

2000). The arguments of unaccusative verbs typically bear a Theme or Patient role (akin to the object of a transitive verb), whereas those of unergative verbs bear an Agent role (like the subject of a transitive verb). This division aligns with the Uniformity of Theta Assignment Hypothesis (UTAH) (M. Baker, 1988, p. 46), which states the following:

(3) Uniformity of Theta Assignment Hypothesis:

Identical thematic relationships between items are represented by identical structural relationships between those items at the level of D-structure.

Within the Minimalist Program (Chomsky, 1995), where D-structure is no longer assumed, UTAH is reinterpreted as the principle that a given semantic role is assigned to a consistent structural position. From this perspective, the Agent role is always associated with an external argument merged outside the VP, whereas the Theme/Patient role is assigned to an internal argument merged inside the VP.

3. L2 Acquisition of Unaccusativity

3.1. Unaccusative Trap Hypothesis (Oshita, 1997, 2001)

This section discusses a well-known hypothesis proposed by Oshita (1997, 2001) to explain various phenomena related to unaccusativity in L2 acquisition and the developmental trajectory in this area. Oshita (2001) shed critical light on phenomena found in L2 English, including ungrammatical passive unaccusatives and learners' sensitivity to the unaccusative–unergative distinction, and proposed the Unaccusative Trap Hypothesis (UTH), which elegantly accounts for interesting phenomena concerning L2 unaccusativity. The UTH predicts that L2 learners undergo three developmental stages in the acquisition of unaccusativity: lexical confusion, syntactic confusion, and full acquisition.

In the initial stage, L2 learners are insensitive to the argument structure distinction between unaccusatives and unergatives, resulting in their interlanguage grammar treating unergatives and unaccusatives as having an external argument, in violation of UTAH. Oshita (1997, 2000a, 2001) conjectures that this initial confusion may stem from the canonical NP-V word order in the input, which encourages learners to associate the single argument of intransitive verbs with an external argument. The misanalysis of the

argument structure of unaccusatives would paradoxically allow learners to produce and accept grammatical active sentences with unaccusatives. Notably, the UTH assumes that despite their insensitivity to the unaccusative–unergative contrast in argument structure, “learners are believed to be aware of general semantic differences among intransitive verbs” (Oshita, 2000a, p. 245). Learners, even in the first stage of UTH, are sensitive to the unaccusative–unergative distinction at the level of lexical semantics. In the second stage, learners realize the target argument structure in which unergatives and unaccusatives have external and internal arguments, respectively. Subsequently, the target argument representation in learners’ interlanguage grammar poses another problem—non-target passivization of unaccusatives, as detailed in Section 3.2. Learners make passive errors exclusively with unaccusative verbs by overgeneralizing the overt marking of NP-movement in passive structures to unaccusative structures. The target argument realization also causes learners’ reluctance to accept unaccusatives in the NP-V order, a phenomenon documented in previous studies (Kellerman, 1978; Oshita, 2002, 2004). In the final stage, learners fully acquire argument structures and syntactic derivations of unaccusatives, eliminating passive errors. The transition from the second to the final stage is achieved through indirect negative evidence in the input. They are repeatedly exposed to unaccusative verbs in the target NP-V order, which they are urged to passivize. According to Oshita (2001), as learners’ non-target passivization of unaccusatives occurs because of the overt marking of NP-movement with passive morphosyntax, repeated exposure to correct unaccusative sentences helps learners realize that the object-to-subject movement of NPs in the derivation of unaccusatives does not involve passive morphosyntax. These three stages are summarized in Table 1 below.

Table 1

Summary of the Three Developmental Stages of the UTH

Stage	Verb	Argument structure	Syntactic structure
1st	Unergative	(x)	NP-V
	Unaccusative	(*x)	NP-V
2nd	Unergative	(x)	NP-V
	Unaccusative	<y>	*NP- <i>be-Ven</i>
3rd	Unergative	(x)	NP-V

Unaccusative	<y>	NP-V
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Here, “x” and “y” stand for an external argument and an internal argument, respectively. As shown in Table 1, learners at the first stage misanalyze an unaccusative argument as an external argument; however, consequently, they produce seemingly correct unaccusative sentences in the NP-V order. Once they achieve the final stage, non-target passivization does not occur because of the full acquisition of unaccusativity.

Throughout these three stages, a U-shaped pattern in the development of the L2 acquisition of unaccusativity is expected to emerge with respect to passivization errors (Oshita, 1997, 2001, 2014, 2022). Learners do not make errors with unaccusatives in the first stage, start to make passive errors in the second stage, and recover from these errors in the final stage. Contrary to this prediction, previous studies attempting to find a U-shaped pattern have failed to observe it (e.g., Deguchi & Oshita, 2004; Oshita, 1997; Park & Lakshmanan, 2007; Yamakawa et al., 2003). Based on these findings, Oshita (2014) distinguished between absolute and relative U-shapes, noting that the UTH does not predict absolute U-shaped development but relative U-shaped development. In his definition of an absolute U-shape, more passive errors with unaccusatives are found in the middle stage of development than in the initial or final stages. However, in the definition of the relative U-shape, passive unaccusatives appear more frequently than passive unergatives in the middle, but differences in passive error rates between unaccusatives and unergatives are not observed in the initial and final stages. Under the notion of a relative U-shaped curve, learners in the middle stage do not become worse when judging passive unaccusatives than when they are in the initial stage. Nevertheless, their judgment of passive unaccusatives deteriorates compared to that of passive unergatives. The UTH’s prediction was confirmed in Oshita’s (2014) cross-sectional study. He observed a clear U-shaped development in which L2 English learners judged passive unaccusatives less correctly than passive unergatives at the middle-proficiency level.

3.2. Passivization errors

L2 learners of English frequently produce and accept ungrammatical passive constructions involving intransitive verbs, as shown in (4).

- (4) a. *He is also appeared on the list of investigations of gold smuggling.
 b. *I was nearly arrived to my office. (a–b from Oshita, 2000, p. 314)
 c. *The most memorable experience of my life was happened 15 years ago.
 d. *My mother was died when I was just a baby. (c–d from Zobl, 1989, p. 202)

Such errors are predominantly observed with unaccusative verbs but rarely with unergative verbs (Oshita, 2000, 2001; Tazaki, 2024b). Moreover, as these errors are found across various L1 backgrounds, this phenomenon is considered a universal error (Ju, 2000). Over the past few decades, many researchers have proposed competing explanations to uncover the root cause of these errors (see also Wakabayashi, 2025, for an overview of existing explanations). This section reviews three major analyses: discourse-pragmatics (Ju, 2000), subject animacy (Chung, 2014), and the overt marking of NP-movement (Oshita, 2000). Sections 3.2.1, 3.2.2, and 3.2.3 examine each analysis and highlight the challenges.

3.2.1. Conceptualizable agent analysis (Ju, 2000)

We begin with a discourse-pragmatic account of passivization errors. According to Ju (2000), passive unaccusatives are more likely to occur when the context implies a conceptualizable agent (CA). When an agent-like entity can be inferred from the surrounding discourse, L2 learners are more inclined to passivize unaccusative verbs. To illustrate this, compare the contexts in (5) and (6), taken from Ju (2000, pp. 109–111):

- (5) The police were called in to remove a strange package.

The package disappeared immediately.

- (6) A car slid off the road and a package fell from the car into the lake.

The package disappeared immediately.

Although both examples include the same unaccusative sentence—*The package disappeared immediately*—the preceding context influences its interpretation. In (5), the police serve as a plausible external causer of the disappearance event. Conversely, no

such agent is evident in (6). Thus, the disappearing event is perceived as being caused externally in (5) and internally in (6). Building on this discourse distinction, Ju predicted that L2 learners would be more likely to apply passive morphology to unaccusatives in contexts such as (5), in which a CA is implied, than in contexts such as (6), in which it is not.

Ju tested this prediction using a forced-choice task in which 35 Chinese learners of English read pairs of sentences and chose between the active and passive forms for the target sentence, as illustrated in (7):

(7) Mary bought some fish at the market.

It (prepared / was prepared) soon after.

The materials included 13 alternating unaccusative verbs and five non-alternating unaccusatives, each presented in two discourse contexts: one implying external causation (as in (5)) and one implying internal causation (as in (6)). Ju predicted that participants would prefer passive forms in externally caused contexts and active forms in internally caused contexts. This prediction was confirmed by the results.

The study revealed a significant difference: passive responses were chosen more often in externally caused contexts than in internally caused contexts. Similar findings have been reported for learners from other L1 backgrounds, including Chinese (Chung, 2014), Iranian (Sahragard et al., 2010), Japanese (Okamura & Shirahata, 2024b), and Korean (Choi, 2019; Chung, 2014; Kim et al., 2025).

Despite this cross-linguistic support, the CA analysis faces criticism. As Oshita et al. (2019) point out, Ju's analysis is grounded in the Transitivity Hypothesis (Yip, 1995), which claims that passive unaccusatives result from the semantic passivization of lexically causativized unaccusative verbs. According to this view, when a CA is present, an external argument is added to the argument structure of an unaccusative verb and suppressed via passivization. This hypothesis predicts that L2 learners should also produce ungrammatical lexically causativized unaccusatives, such as the following:

(8) a. * The shortage of fuels occurred the need for economical engine.

b. *Careless currency devaluation will go back us to old habit.

- c. * This construction will progress my country.
(a– c from Rutherford, 1987, p. 89)
- d. *... he falls a piece of note into dough by mistake.
- e. * I rose my hand.
(d–e from Oshita, 2000. p, 313)

Such errors have been documented (e.g., Cabrera & Zubizarreta, 2005; Hirakawa, 1995, 2000; Matsunaga, 2005; Oshita, 1997, 2000; Tazaki, 2024a, 2025). However, they are relatively rare, especially when compared to passive unaccusatives (see Oshita, 2000; Tazaki, 2024a, 2024b). This poses a challenge to the Transitivity Hypothesis and, by extension, to Ju's CA analysis.

Critiques of the CA analysis and the Transitivity Hypothesis have been developed in detail elsewhere (see Oshita, 2000, 2022). In summary, although the CA account assumes that unaccusatives are lexically causativized in the L2 grammar, the empirical evidence suggests that learners seldom produce such constructions.

Despite these shortcomings, Ju's central insight—that L2 learners are sensitive to the discourse-pragmatic context when producing passive unaccusatives—remains valuable. This study preserves this insight while providing an alternative explanation that does not require the assumption of the lexical causativization of unaccusatives.

3.2.2. Subject animacy analysis

Another account of passive unaccusatives focuses on the effect of subject animacy (e.g., Chung, 2014; Shirahata et al., 2019; No & Chung, 2006; Oh, 2014; Okamura & Shirahata, 2024a, b). This account predicts that L2 learners are more likely to produce and accept passive unaccusatives when the subject is inanimate rather than animate. The underlying rationale is that learners associate the subject and object of transitive verbs with the Agent and Theme roles, respectively, and consequently interpret the subject of an unaccusative verb as a derived subject, akin to the surface subject in a passive construction (Chung, 2014). Thus, when the unaccusative subject is inanimate, learners may be more inclined to passivize the sentence. This prediction is supported by data from learner production (Oh, 2014; Tazaki, 2024b), acceptability judgment tasks (No & Chung, 2006; Otaki &

Shirahata, 2017), and forced-choice experiments (Chung, 2014).¹ Tazaki (2024b) used a corpus of Japanese learners to examine how often passive unaccusatives occurred with animate and inanimate subjects. The results showed that errors were more frequent with inanimate subjects. Similarly, Chung (2014) conducted a forced-choice task with Korean and Chinese L2 English learners and found that passive responses were significantly more common when the subject was inanimate.

Although several studies support the animacy account, its status as the primary cause of passive unaccusatives remains questionable. One key problem is that some unaccusative verbs, such as *die*, typically take animate subjects yet frequently appear in erroneous passive constructions (see Appendix A in Oshita, 1997 and Tazaki, 2024c, for the frequency of passive errors with *die*). According to the animacy account, learners should not produce such errors with animate subjects, making it difficult to explain the high rate of passive errors with verbs such as *die*.

Moreover, similar to the CA analysis, the animacy-based account implicitly assumes the lexical causativization of unaccusatives. Chung (2014, p. 63) states that “...L2 learners would make more errors with non-animate subjects than with animate subjects since the subject of an unaccusative would be considered to be a Theme subject of a passive.” Otaki and Shirahata (2017, p. 189) also say that “[t]hey [= learners] may have thought that the nouns [= non-animate subject nouns of intransitive verbs] should be required to have an external subject.” This assumption parallels that of the CA analysis and therefore inherits its limitations (see Section 3.2.1; see also Tazaki, 2024c, for additional critiques).

3.2.3. Non-target overt marking of NP-movement

A third account of passive unaccusatives is the overt marking of NP-movement (Deguchi & Oshita, 2004; Oshita, 1997, 2000b, 2001, 2014, 2022; Zobl, 1989). According to this

¹ One might wonder if the findings are expected, as English learners are often taught that inanimate subjects appear in passive sentences. While this possibility cannot be dismissed, it appears implausible that all learners who generate passive unaccusatives with inanimate subjects have explicitly been taught this pattern; thus, I assume that the frequency of passive unaccusatives with inanimate subjects is not significantly influenced by instruction associating inanimate subjects with passive constructions. I am thankful for Hiroyuki Shimada (p.c., 2025) for bring up the possibility to me.

view, learners overgeneralize passive morphology (*be* + *-en*) to unaccusative constructions because of the strong association between passive syntax and NP-movement. As illustrated in (9) and (10), passive and unaccusative sentences involve NP-movement of the Theme argument from an internal position to the subject position (Spec-TP):

(9) Passive verb

- a. Leslie kissed Brian.
- b. Brian_i was kissed *t_i* by Leslie.

(10) Unaccusative verb

- a. There arrived three boys at the school.
- b. The three boys_i arrived *t_i* at the school.

Owing to this structural similarity, learners may mistakenly assume that passive morphology is required when NP-movement occurs, including in unaccusative constructions. A key assumption is that learners can distinguish between unaccusative and unergative verbs in terms of argument structure and underlying syntax. Learners who produce passive errors with unaccusatives are considered to be in the middle of the three developmental stages of the acquisition of unaccusativity, as discussed in Section 3.1.

Unlike the CA and animacy accounts, this hypothesis does not assume semantic passivization or lexical causativization of unaccusatives. As such, it does not predict the use of causative constructions and can account for passive errors with animate subjects, which the animacy account fails to explain. As unaccusatives always involve a single internal argument—regardless of the subject’s animacy—this approach treats the passive overgeneralization as a syntactic misanalysis, not a semantic one.

Nevertheless, the NP-movement account does not deny the influence of discourse-pragmatic context (CA) or subject animacy. To unify these perspectives, the present study proposes a new framework, *the Theta Misanalysis Hypothesis*, which builds on the NP-movement account while integrating semantic and discourse-based factors.

4. Proposal

4.1. Theta Misanalysis Hypothesis

The three previous accounts—CA, subject animacy, and NP-movement—highlight different aspects of L2 learners’ passivization errors. However, none of them provides a unified explanation that accounts for syntactic, discourse-pragmatic, and semantic factors. To address this gap, I propose a new hypothesis: *the Theta Misanalysis Hypothesis*, which builds on the NP-movement account while incorporating insights from CA and subject animacy analyses. It centers on the idea that learners misanalyze the theta role of the subject in unaccusative constructions, which leads to the inappropriate application of passive morphology.

Thus far, we have established that unaccusatives are prone to passivization because their sole argument is base-generated as an internal argument, and that passive unaccusatives result from learners’ strong association of passive morphosyntax with NP-movement. The syntactic base position of an argument is a crucial factor in determining whether an intransitive verb is susceptible to non-target passivization. The UTAH predicts that Theme/Patient arguments must be generated inside the VP (as internal arguments), whereas Agent arguments must be generated outside the VP (as external arguments). If UTAH holds true, then the theta role assigned to the sole argument is the primary factor influencing passive intransitives. If the argument is analyzed as an Agent in the learner’s interlanguage grammar, it is treated as external; if it is analyzed as a Theme/Patient, it is treated as internal. Assuming that L2 learners are sensitive to the mapping between theta roles and syntactic positions, as UTAH predicts, achieving target-like theta assignment is essential for acquiring unaccusativity. Incorrect or non-target theta role assignment disrupts syntactic realization because different roles correspond to different structural positions.

Dowty (1991, p. 571) proposes that thematic roles are “simply not discrete categories at all, but rather are cluster concepts.” An argument that possesses more Proto-Agent properties than Proto-Patient properties is classified as a Proto-Agent, and vice versa. The lists of Proto-Agent and Proto-Patient properties are listed in (11) and (12), respectively.

(11) *Proto-Agent properties*²

- a. Volition (V)
- b. Sentience (S)
- c. Causing an event (E)

(12) *Proto-Patient Properties*³

- a. Change of state/location (C)
- b. Incremental theme (I)
- c. Causally affected (A)

Under Dowty's Proto-Role Hypothesis, we predict that unaccusative arguments with a higher number of Proto-Patient properties and fewer Proto-Agent properties are more susceptible to passivization errors. This prediction aligns naturally with UTAH's claim that Patient arguments are generated in the VP as internal arguments. Consider the contrast between the unaccusative verb *fall* and the unergative verb *run*. The verb *fall* denotes a change of location and its argument carries the Proto-Patient feature [+C] but no Proto-Agent properties. Therefore, its argument is projected as an internal argument. Conversely, *run* is a manner-of-motion verb whose argument bears Proto-Agent properties [+V, +S] but no Proto-Patient features and is, therefore, projected as an external

² Dowty (1991) included "movement" in the list of Proto-Agent properties to explain the observation that some verbs such as *hit* can have the NP with a "movement" property in the indirect object position, as in *Miki hit the fence with the stick*. In his analysis, a non-subject argument of a three-predicate having a greater number of Proto-Patient properties must be realized as the direct object. If "movement" is a property of Patient Proto-Role, Dowty (1991, p. 595) says, *hit the fence with the stick* must be unacceptable because the prepositional argument, *the stick*, has more Proto-Patient properties than the direct object *the fence*. However, if this is true, an argument of change-of-location intransitives such as *fall* may have a Proto-Agent role (Dowty, 1991, p. 573), which appears to be questionable because this kind of verb is often diagnosed with unaccusatives in many languages (see J. Baker, 2019, 2020; Sorace, 2000). Therefore, I excluded "movement" from the list of the Proto-Agent properties (11).

³ In the list of the Proto-Patient properties, I excluded "stationary relative to another participant" (Dowty, 1991, p. 572) because it is correspondence of "movement" property, which I excluded for the reason described in footnote 2.

argument. As only internal arguments undergo object-to-subject movement, unaccusatives—but not unergatives—are susceptible to non-target passivization.

Let us now revisit an earlier example involving the unaccusative verb *disappear* in externally and internally caused events (see examples (13) and (14) below).

(13) The police were called in to remove a strange package.

The package disappeared immediately.

(14) A car slid off the road and a package fell from the car into the lake.

The package disappeared immediately.

As *disappear* is a change-of-state verb, its argument inherently bears [+C]. In (13), the noun phrase *a strange package* in the first sentence receives the Proto-Patient property [+A] from the verb *remove*, and this property is “inherited” by *the package* in the following sentence. Consequently, the argument of *disappear* in (13) has two Proto-Patient properties: [+C, +A]. Conversely, in (14), no [+A] feature is inherited from the context; therefore, the argument only has [+C]. As the argument in (13) exhibits a richer Proto-Patient profile than in (14), this explains why learners are more likely to passivize *disappear* in externally caused events. Crucially, this explanation does not assume lexical causativization, distinguishing it from the CA analysis.

This analysis also extends naturally to explain the subject animacy effects. When the subject of an unaccusative verb is inanimate, as in (15a), it typically bears only the property [+C], whereas when it is animate, as in (15b), it bears [+C, (+V), +S].

- | | | | | |
|------|----|---------------------|----------------|---------------------|
| (15) | a. | The letter arrived. | [+C] | (inanimate subject) |
| | b. | Miki arrived. | [+C, (+V), +S] | (animate subject) |

Under Dowty’s Proto-Role Hypothesis, the argument of *arrive* consistently bears [+C], regardless of animacy. However, in L2 learners’ interlanguage grammar, the animate subject in (15b) may be misinterpreted as possessing Proto-Agent properties due to its semantic features. As a result, the argument is misclassified as an external argument, and the sentence is not perceived as eligible for passivization. In contrast, inanimate arguments like in (15a) are more likely to be interpreted correctly as internal arguments

and are thus more prone to passive overgeneralization. This account successfully explains the observed animacy effect in L2 passivization errors.

To summarize, the Theta Misanalysis Hypothesis posits that L2 learners rely on the Proto-Role properties of a single intransitive argument to determine whether it is an internal or external argument. Their theta-role assignment is influenced by the discourse-pragmatic context and subject animacy (agentivity), which in turn affects their use of passive morphology with intransitive verbs. Importantly, this hypothesis builds on the overt marking of NP-movement account, which posits that passive unaccusatives are derived from learners' association of NP-movement with passive morphosyntax. That is, we assume that non-target passivization is structural passivization rather than semantic passivization. Although misanalysis of theta roles may begin at very early stages of L2 development, learners' interlanguage grammar does not exhibit passivization errors with unaccusatives until they reach a later stage—when they begin to distinguish between intransitive verbs that take external arguments (unergatives) and those that take internal arguments (unaccusatives) (see Oshita, 2001). Without awareness of this distinction, the single argument of an intransitive verb is uniformly treated as external, regardless of its Proto-Agent or Proto-Patient properties, rendering discourse and animacy effects invisible.

4.2. Passive Unergative

In Section 4.1, a novel hypothesis—the Theta Misanalysis Hypothesis—was introduced to explain the effects of discourse-pragmatics and subject animacy on passive unaccusatives. Although this hypothesis successfully captures these two phenomena, the passivization of unergative verbs has received comparatively little attention in the literature. This section explores the predictions of the hypothesis regarding this error type.

Although unaccusative verbs have been widely reported to undergo passivization, the passivization of unergatives has received relatively less attention (see Yusa, 2003; Wakabayashi, 2023, for some exceptions). To the best of my knowledge, the influence of discourse-pragmatic contexts on the passivization of unergatives remains largely unexamined. If discourse-pragmatic contexts that facilitate the passivization of unaccusatives influence passivization errors in general, similar contextual effects should

be observed with unergatives. Consider the unergative verb *cry* in the two distinct contexts in (16).

- (16) a. Miki had been practicing hard to win the basketball tournament. However, she couldn't win yesterday's tournament, and she was so frustrated that she couldn't sleep last night.

Miki cried.

- b. Two days ago, Ken and Miki were having an argument. Ken started to get really irritated, and in the end, he hit Miki.

Miki cried.

In (16a), where *cry* occurs in a context devoid of an apparent external cause, the subject possesses only a Proto-Agent property [+S], making it unambiguously an external argument. Conversely, in (16b), the crying event is triggered by a salient external causer *Ken*, and consequently, the subject *Miki* acquires the Proto-Agent property [+S] and the contextually assigned Proto-Patient property [+A]. In the latter scenario, the subject could, in principle, be generated in an external or internal argument position, depending on the L2 learner's interlanguage grammar. Although the exact mechanism that determines the base-generated position of the subject in (16b) remains unclear, the key prediction of the Theta Misanalysis Hypothesis is that unergatives should not be passivized in the absence of an external cause but may be passivized in contexts where such a cause is present. Accordingly, the hypothesis predicts differential error rates between these two contexts.

Interestingly, a similar prediction is obtained from the CA analysis. However, the mechanisms underlying the passivization of unergatives differ significantly between the two accounts. The CA analysis posits that when a CA is present in the discourse, a new external argument is introduced into the argument structure of the unergative verb, demoting the original external argument to internal argument status (see Oshita, 2022). By contrast, the Theta Misanalysis Hypothesis does not assume the addition of an external argument; instead, it holds that the sole argument of the unergative verb is analyzed as an internal argument in argument structure.

Owing to this fundamental difference, the two analyses make opposing predictions regarding lexical causativization. The CA analysis predicts that unaccusatives

and unergatives should exhibit lexical causative errors, as it assumes that passivization results from the lexical causativization of intransitive verbs. Conversely, the Theta Misanalysis Hypothesis does not predict causative errors with unergatives because it does not involve lexical causativization. Therefore, if causativized unergatives are largely absent from L2 learner data, this would challenge the CA account. Contrarily, such an absence would pose no issue for the Theta Misanalysis Hypothesis. Indeed, as discussed in Section 4.2.1, causative unergatives have been sparsely documented in the literature, contrary to the more frequent documentation of passivized and causativized unaccusatives.

5. Experiment: Discourse-Pragmatic Effects on Passive Intransitives

This experiment primarily examined the effects of discourse-pragmatic factors on the passivization of unergative verbs. Specifically, this study investigated whether contextual factors influence the likelihood of non-target-like passivization of unergatives.

5.1. Methods

5.1.1. Participants

The participants were L1 Japanese-speaking learners of L2 English whose TOEIC scores ranged from 550 to 780, corresponding to CEFR level B1. This proficiency range was chosen based on prior studies indicating that learners at the B1 level exhibit sensitivity to the unaccusative–unergative distinction (Tazaki, 2024a; Tazaki, 2025). Learners' sensitivity to this distinction is crucial for testing the Theta Misanalysis Hypothesis. The hypothesis predicts contextual effects only if learners' interlanguage grammar distinguishes between intransitive verbs that take external arguments and those that take internal arguments. Previous studies that conducted an acceptability judgment task have shown that English learners whose TOEIC scores fall between 550 and 780 rate passive unaccusatives higher than passive unergatives, indicating their sensitivity to the unaccusative–unergative difference.

Participants were recruited through the Japanese crowdsourcing platforms CrowdWorks and Lancers. Only data from participants who had taken the TOEIC within one year prior to the experiment were included. A total of 60 participants were recruited for this study. However, data from 31 participants were excluded based on the following

criteria: (i) failure to demonstrate knowledge of passive constructions (17 participants)⁴ (see Section 5.1.3); (ii) lack of self-reported TOEIC score (five participants); (iii) TOEIC score obtained more than one year before participation (four participants); or (iv) score outside the target range (five participants). The final analysis included data from 29 participants.

As crowdsourcing has been employed less frequently to recruit participants in L2 research (especially in studies that conduct acceptability judgment tasks), this section briefly discusses the advantages and disadvantages of using crowdsourcing platforms for L2 research. In L1 research, several studies have suggested that adult L1 acceptability judgment data obtained through crowdsourcing are comparable to data collected in laboratory settings (e.g., Sprouse, 2011). However, to the best of our knowledge, no study has investigated the reliability of L2 acceptability judgment data obtained via crowdsourcing. One might question the quality of such data, specifically, whether the data collected online are comparable to those collected in face-to-face settings. As some participants fail to engage seriously in tasks even in laboratory experiments, the use of crowdsourcing does not appear to be inherently problematic (Nakatani, 2024). Moreover, some studies have shown that data obtained from online and laboratory experiments are compatible (e.g., Ishihara & Hamada, 2025).

Furthermore, as demonstrated in this study, the results of our experiment closely align with findings from previous research, which have consistently shown L2 learners' sensitivity to the unaccusative–unergative distinction (e.g., Kondo, 2005; Matsunaga, 2005; see also Tazaki, 2025, which used crowdsourcing service to recruit participants and found similar results). The replication of these results suggests that crowdsourcing can serve as a viable and reliable method for conducting acceptability judgment tasks in L2 research. Additionally, participants are less likely to feel nervous, as they can complete the task in a relaxed environment. In the face-to-face experiment, the experimenters are

⁴ The relatively large number of participants who failed to accept grammatical passive sentences may be attributable to the high proportion of ungrammatical passive sentences in the task. As participants were exposed to many ungrammatical passives, this exposure may have led them to reject grammatical passive sentences as well. A detailed investigation of the acquisition of the passive construction is beyond the scope of this study, and we leave this issue for future research.

typically in the same room in which the participants are performing the task. These situations may have imposed stress on the participants. However, in the online setting, participants were free from the stress of being kept under observation.⁵

5.1.2. Materials and procedure

The experiment employed an acceptability judgment task that included active (NP-V) and passive (NP-V.p.p.) forms of unaccusative and unergative verbs. Each target sentence was preceded by a short contextual paragraph in Japanese to ensure complete comprehension. Twelve target verbs were selected and divided equally across three lexical categories: (1) change-of-state/location (*die, appear, disappear, fall*); (2) non-controlled processes (*cry, laugh, smile, sleep*); and (3) controlled processes (*walk, dance, sing, study*). The first category represents unaccusatives; the latter two represent unergatives.

Each verb appeared in the active and passive structures and was counterbalanced across participants. Thus, each verb was presented twice: in an active sentence with a CA and in a passive sentence without one (Group A) or in an active sentence without a CA and in a passive sentence with one (Group B). This design minimized task length while enabling within-subject comparison of contextual conditions. Table 2 illustrates the distribution of target verbs across groups (“IC” and “EC” denote “Internal Causation” and “External Causation,” respectively).

Table 2

Distribution of target verbs

Group	Verb	Causation	Structure
A	<i>a</i>	IC	NP- <i>a</i> .
		EC	*NP- <i>be-a_{p.p.}</i> .

⁵ However, one major concern is the possibility that participants may provide false information in order to receive compensation. In our study, we targeted only participants who had taken the TOEIC test. In this situation, these participants are motivated to pretend to have their scores for monetary reasons. Admittedly, we have no way to determine whether any participants provided false information. Nevertheless, in our study, participants are sensitive to the unaccusative–unergative distinction, and they indeed show their sensitivity (see Section 5.3).

	<i>b</i>	IC	*NP- <i>be-b</i> _{p.p.}
		EC	NP- <i>b</i> .
	<i>c</i>	IC	NP- <i>c</i> .
		EC	*NP- <i>be-c</i> _{p.p.}
	<i>d</i>	IC	*NP- <i>be-d</i> _{p.p.}
		EC	NP- <i>d</i> .
	<i>a</i>	IC	*NP- <i>be-a</i> _{p.p.}
		EC	NP- <i>a</i> .
B	<i>b</i>	IC	NP- <i>b</i> .
		EC	*NP- <i>be-b</i> _{p.p.}
	<i>c</i>	IC	*NP- <i>be-c</i> _{p.p.}
		EC	NP- <i>c</i> .
	<i>d</i>	IC	NP- <i>d</i> .
		EC	*NP- <i>be-d</i> _{p.p.}

For example, if a participant was assigned to Group A and the verbs a–d corresponded to *die*, *appear*, *disappear*, and *fall*, then *die* and *disappear* would appear in an active structure without a CA and in a passive structure with a CA. Conversely, *appear* and *fall* would appear in a passive structure with a CA and an active structure without a CA.

This experiment included two different contexts—internal causation and external causation—because the Theta Misanalysis Hypothesis expects causation to influence passive intransitives irrespective of verb type. None of the previous studies has yet examined the contextual effects on passive unergatives. Therefore, this study examines the effects on unergatives.

Each participant evaluated 24 experimental sentences. Additionally, 15 filler items were included, for a total of 39 sentences per participant. (See the Appendix for a full list of target sentences.)

To control for subject animacy, all unaccusative subjects were inanimate, while all unergative subjects were animate. This choice was motivated by the desire to exclude the possibility that L2 learners would analyze the agentive animate subjects of

unaccusatives as external arguments (Tazaki, 2025c).⁶ To isolate discourse effects, only inanimate subjects were used in unaccusative conditions. Examples of the target sentences are provided below (target sentences are italicized):

(17) Unaccusative

a. Internal causation

Emily’s ring slipped off her finger and rolled down the slope. Unfortunately, the ring sank into the huge lake.

The ring disappeared / was disappeared into the huge lake.

b. External causation

The magician tightly clenched the coin in his right hand. Then, he struck his right hand, which held the coin, with his left hand.

The coin disappeared / was disappeared from his right hand.

Participants rated each sentence on a 7-point scale: 1 for “incorrect,” 2 for “probably incorrect,” 3 for “may be incorrect,” 4 for “cannot decide,” 5 for “may be correct,” 6 for “probably correct,” and 7 for “correct.” Before the main task, the participants completed a practice session using the same rating scale.

Investigating the discourse-pragmatic effects on passive intransitives is meaningful only if learners have already acquired passivization. To ensure that the

⁶ An anonymous reviewer raises the question of whether problems arise if animate subjects are used uniformly across verb classes. A potential interpretive problem arises if participants judge all classes of intransitive verbs in the passive construction in the same way—for example, if they reject all passivized intransitives. In such a case, the results can be interpreted in two ways: (i) participants are insensitive to the unaccusative–unergative distinction at the level of argument structure, or (ii) participants are sensitive to the distinction, but due to subject animacy, passive unaccusatives are also rejected.

To test the predictions of the Theta Misanalysis Hypothesis, it is crucial to ensure that participants are sensitive to the unaccusative–unergative dichotomy. By using prototypical inanimate subjects with unaccusative verbs, we can avoid the potential problem described above. If participants assign similar ratings to all types of intransitive verbs, this suggests that they are insensitive to the distinction between unergatives and unaccusatives and are, therefore, not suitable participants for this study. Conversely, if they accept passive unaccusatives more readily than passive unergatives, they can be considered suitable participants, as this indicates sensitivity to the distinction.

participants knew passivization, five grammatical passive sentences were included as fillers. These also served as a pre-test. Participants who rated at least four out of five of these sentences as 5 or above were included in the final analysis.

To confirm the naturalness of the stimuli, two native speakers of English reviewed all target and filler items.

5.2. Predictions

Based on previous studies, L2 English learners are expected to judge passive unaccusatives as acceptable more frequently in externally caused events than in internally caused events. This expectation is consistent with the Theta Misanalysis Hypothesis. If this hypothesis is correct, then a similar pattern should emerge for passive unergatives. Notably, the Theta Misanalysis Hypothesis predicts the difference between the two unergative types, contrary to other existing analyses.

To refine these predictions, we examine the Proto-Role properties associated with each lexical category under two causation conditions. Table 3 summarizes these properties, where “CoS/L,” “NCP,” and “CP” refer to change-of-state/location, non-controlled process, and controlled process, respectively. The rows labelled “Agent” and “Patient” indicate the Proto-Role properties assigned to the sole argument of each verb type. *x* and *y* represent external and internal arguments, respectively.

When a CoS/L verb appears, regardless of whether the event denoted by the verb is internally caused or externally caused, it has only Proto-Patient properties; therefore, its argument is projected as an internal argument. As an intransitive verb with an internal argument is passivized, this class of verbs is predicted to be accepted in passive constructions. NCP verbs bear the Proto-Agent property [+S]; thus, their argument is projected as an external argument. When they occur in the externally caused event, their argument also bears a Proto-Patient property [+A]. This contextual effect allows arguments to be generated either inside or outside a VP. The difference between the two conditions in the same verb class, NCP verbs, is predicted to be that those in the externally caused event are more acceptable than those in the internally caused event. As for CP verbs, in either context, they have two Proto-Agent properties, rendering their argument external. Although their argument is always external, passivized CP verbs would be more

acceptable in externally caused events than in internally caused events because of the presence of the Proto-Patient property [+A] in the former.

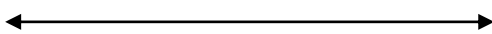
Table 3

Proto-Role properties for each lexical category and causation

Lexical item	Causation	Agent	Patient	Argument
CoS/L	IC	∅	+C	<y>
	EC	∅	+C, +A	<y>
NCP	IC	+S	∅	(x)
	EC	+S	+A	(x) or <y>
CP	IC	+V, +S	∅	(x)
	EC	+V, +S	+A	(x)

Given the distribution of Proto-Agent and Proto-Patient properties across lexical categories, the predicted acceptability of passive intransitives follows the ranking in (18), where “A > B” indicates that A is more likely than B to be judged acceptable:

$$(18) \quad \text{CoS/L} > \text{NCP} > \text{CP}$$



Acceptable
Unacceptable

The sole argument of CoS/L verbs bears one or more Proto-Patient properties and no Proto-Agent properties. In contrast, the argument of NCP verbs bears one Proto-Agent and, under external causation, one Proto-Patient property. The argument of CP verbs has the highest number of Proto-Agent properties, making it the most resistant to passive interpretation. These differences in Proto-Role properties form the basis for the predicted acceptability patterns in L2 grammar.

As many previous studies have observed that unaccusatives with inanimate subjects are prone to passivation, a contrast in acceptability between unaccusatives (i.e., CoS/L) and unergatives (i.e., NCP and CP) is expected. More importantly, the Theta Misanalysis Hypothesis predicts the difference between CoS/L and NCP/CP and that between NCP and CP verbs.

interaction between VERB TYPE and CAUSATION was also not significant ($p > .05$), suggesting that the effect of VERB TYPE remained stable across different contexts.

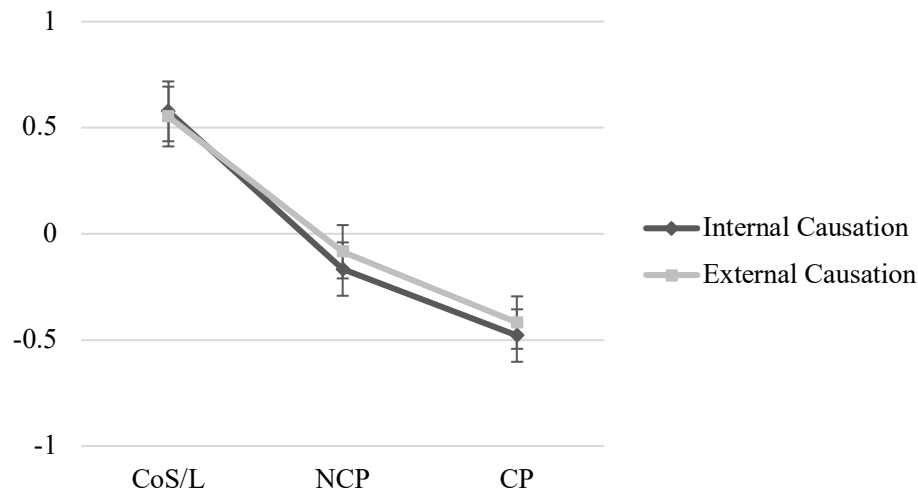


Figure 1

Mean z-scores for three lexical classes and two causation conditions

5.4. Discussion

The findings confirm that L2 learners are more likely to accept passive unaccusatives (CoS/L verbs) than passive unergatives (NCP and CP verbs). More crucially, there are differences in acceptability between unergative classes: NCP and CP verbs. NCP verbs are more acceptable in passive constructions than CP verbs. This finding supports the hypothesis that learners are sensitive to the Proto-Role properties of the sole argument. These results align with prediction (18), repeated as follows:

$$\begin{array}{ccccc}
 (20) & \text{CoS/L} & > & \text{NCP} & > & \text{CP} \\
 & \longleftarrow & & \longrightarrow & & \\
 & \text{Acceptable} & & & & \text{Unacceptable}
 \end{array}$$

The observed gradation in the acceptability judgments of passive intransitives aligns with the predictions. CoS/L verbs possess only Proto-Patient properties, whereas NCP verbs exhibit more Proto-Agent properties than CoS/L verbs but fewer than CP verbs, which bear the highest number of Proto-Agent properties. This variation in Proto-Role properties accounts for the observed pattern. The sole argument of CoS/L verbs is

necessarily base-generated inside the VP, whereas that of CP verbs is generated outside the VP. The argument of NCP verbs can be base-generated either inside or outside the VP. As NP-movement occurs only with internal arguments (Oshita, 2000; Tazaki, 2024b), the argument of CoS/L verbs is the least resistant to non-target passivization. In contrast, CP verbs are the most resistant because their arguments are never base-generated as internal arguments in the verb's complement.

As the results show that unaccusatives occurred with inanimate subjects while unergatives occurred with animate subjects, one might wonder if this pattern can be attributed solely to subject animacy. However, this analysis cannot hold because NCP and CP verbs involved animate subjects, but they exhibited different levels of acceptability. The animacy-based account fails to account for the contrast, posing a challenge.

Although the findings of this experiment support the prediction (18/20), a more fine-grained prediction remains unexplained. The expected ranking of acceptability, incorporating causation effects, is presented in (21).

$$(21) \text{ CoS/L+EC} > \text{CoS/L+IC} > \text{NCP+EC} > \text{CP+EC} > \text{NCP+IC} > \text{CP+IC}$$



Acceptable
Unacceptable

As described in Section 5.3, contextual differences did not significantly affect judgments of passive intransitives. This finding poses a challenge for Ju's (2000) CA analysis and, at first glance, the Theta Misanalysis Hypothesis. According to the CA analysis, discourse-pragmatic contexts enable L2 grammar to introduce additional external arguments into the argument structure. From this perspective, passivization errors occur only when an external causer is present in the context. Thus, the absence of contextual effects in this study presents a critical problem. However, the Theta Misanalysis Hypothesis remains compatible with these results, as discussed earlier. As this hypothesis does not assume lexical causativization of intransitive verbs, it predicts that passivization errors can arise independently of contextual information. Accordingly, the lack of contextual effects does not undermine the validity of the Theta Misanalysis Hypothesis.

Before concluding, a possible explanation for the absence of contextual effects warrants further consideration.⁷ One possibility is that the discourse context does not affect Proto-Role properties. Indeed, previous studies have reported mixed findings regarding the role of causation in L2 passivization. Kondo (2005) and Ortega et al. (2018) failed to replicate Ju’s (2000) original findings. Nevertheless, other studies have found supporting evidence for causation effects (see Section 3.2.1). How can these inconsistencies be explained? Although conclusive answers remain elusive, one possibility lies in the developmental trajectory of unaccusativity acquisition. According to the Unaccusative Trap Hypothesis (Oshita, 1997, 2001; see also Section 3.1), L2 learners progress through three stages: an initial stage in which they do not distinguish unaccusatives from unergatives; an intermediate stage in which they begin to distinguish the two, resulting in non-target passivization of unaccusatives; and a final stage in which these errors disappear. The intermediate stage can probably be further divided into two substages: an early second stage, during which learners are sensitive to contextual cues, and a later second stage, during which they no longer rely on such cues but use passive morphosyntax with unaccusatives. If the participants in this study were at the later intermediate stage, their acceptance of passive unaccusatives and rejection of passive unergatives—regardless of discourse context—can be accounted for. However, this hypothesis remains tentative and warrants further investigation.

Finally, this study did not examine other potential factors, such as telicity (Yusa, 2003; Hirakawa, 2006; see also Okamura & Shirahata, 2024b for experimental evidence against telicity effects). Wakabayashi (2023) proposed a novel account in which, in L2 English interlanguage, the passive construction *be* + *en* functions as a “focus” marker, emphasizing the resulting state or endpoint of an event. Under this analysis, intransitive sentences are acceptable in L2 English when a phrase denoting a clear endpoint or location is present, as illustrated in (22).

⁷ One anonymous reviewer suspects that the lack of contextual effects in this study is due to the lack of participants’ attention to contextual sentences. As the task did not include attention-check questions to ensure participants attended to the contextual sentence carefully, this is undeniable. Future studies must address this issue. I am grateful to the reviewer for drawing this point up to me.

- (22) a. *Miki was walked to the village.
 b. *Miki was walked on the treadmill.

According to this proposal, (22a) is acceptable because the goal phrase *to the village* provides a clear resultant state or change in location. In contrast, (22b) is deemed unacceptable because of the absence of a goal phrase that marks such a change. However, this analysis fails to account for the findings of the current experiment, thereby suggesting that the Theta Misanalysis Hypothesis offers a more comprehensive explanation. Moreover, the Theta Misanalysis Hypothesis can incorporate telicity effects without positing a focus-marking function for *be + en*. Specifically, in (22a), the presence of the goal phrase allows the assignment of a Proto-Patient property [+C] to the subject, in addition to its Proto-Agent properties [+V, +S]. This results in an interpretation where *Miki* undergoes a change in location. In contrast, in (22b), where no goal phrase is present, the subject retains only Proto-Agent properties [+V, +S]. Thus, the Theta Misanalysis Hypothesis successfully accounts for the differential acceptability of (22a) and (22b) by appealing to argument structure and Proto-Role properties rather than focus.

6. Implications

The Theta Misanalysis Hypothesis is advanced in this study primarily to provide a unified account of passive unaccusatives; however, as an anonymous reviewer also points out, it can be extended to the more general acquisition of verbs. In this section, we show that the non-target-like anticausativization observed in Tazaki (2026) and the resistance to accepting grammatical anticausative constructions observed in Oshita (2002) can be accounted for by this hypothesis.

Verbs are classified into two types. The first is alternating, which can be used intransitively and transitively, as shown in (23).

- (23) a. Brian broke the car.
 b. The car broke.

The transitive verb *broke* is the typical alternating verb that can be used transitively, as in (23a), and intransitively, as in (23b). The subject of the intransitive use (23b) corresponds to an object in the transitive *break* (23a). As the transitive *break* denotes the causation of the change and its intransitive counterpart denotes only the change without explicitly expressing its causation, the former and latter types are referred to as causative and anticausative verbs, respectively.

The other type is non-alternating verbs, which can be used intransitively or transitively. In (24), the unergative verb *run* occurs in the intransitive construction but not in the causative construction; in (25), the manner transitive verb *kick* occurs in the transitive construction but not in the anticausative construction.

- (24) a. *Leslie ran Kevin. (Intended meaning: Leslie made Kevin run.)
a. Kevin ran

- (25) a. Alec kicked Dylan.
b. *Dylan kicked. (Intended meaning: Dylan was kicked by someone.)

Despite the ongoing debate on explanations of the causative/anticausative alternation, there is a consensus that causation plays an important role (see, e.g., Kageyama, 1996; Levin & Rappaport Hovav, 1995; Reinhart, 2002, 2016). Reinhart (2002, 2016) argues that causative alternation is possible if the verb denotes causation but does not lexically specify the property of the subject. Consider (26) and (27), which were partially adapted from Reinhart (2016, p. 24).

- (26) a. Max / the key / the storm opened the window.
b. The window opened.

- (27) a. The baby / *the spoon / *hunger ate the soup.
b. *The soup ate.

The causative verb *open* allows for agent, instrument, and natural force as its subject, as in (26a), and it has an anticausative counterpart, as in (26b). Conversely, the transitive

verb *eat* requires an agentive subject, as in (27a), and cannot be used in an anticausative construction, as in (27b).

Reinhart's observation and generalization can be restated under the proposed Theta Misanalysis Hypothesis, as follows:

- (28) Only verbs that lexically specify [+C] but not [$\pm V$] and [$\pm S$] can participate in the causative/anticausative alternation.

Let us examine how (26) and (27) can be explained using (28). As shown in (26), *break* does not lexically specify [$\pm V$] and [$\pm S$], and thus, it can be used in the anticausative construction. However, *eat* in (27) requires an animate, volitional subject, indicating the verb specifies [$\pm V$] and [$\pm S$]. Owing to the specification of [$\pm V$] and [$\pm S$], it is not qualified for the causative alternation.⁸

Although the generalization (28) accounts for most causative alternation verbs, some verbs that fall into the generalization do not alternate. For example, the causative verbs *destroy* and *kill* allow various subject types, as shown in (29).

- (29) a. The vandals / the storm / the intense heat destroyed the crops.
b. The marauders / the poison / the cold killed the chickens.

(Rappaport Hovav, 2014, p. 15)

However, these verbs do not allow anticausative uses.

- (30) a. *The crops destroyed.
b. *The chickens killed.

Although verbs that do not follow the generalization (28) do not enter the causative alternation, those that fall into the generalization but fail to alternate vary from language

⁸ This analysis assumes that anticausatives are derived from causative counterparts. Non-alternating anticausatives, such as *fall* and *disappear*, are assumed to be “frozen” in the lexicon, following Reinhart (2016). See also Rappaport Hovav, 2014; Tazaki, 2026, for relevant discussion.

to language. Therefore, L2 English learners cannot rely on their L1 lexical knowledge to determine whether a given verb is alternating or not (see Nagano, 2015; Tazaki, to appear, for relevant discussion).

If L2 learners know the generalization (28) that is formulated under the Proto-Role theory, they are likely to overapply the generalization to non-alternating causative verbs like (29). Moreover, it is predicted that they do not overgeneralize the causative alternation to verbs that lexically specify $[\pm V]$ and $[\pm S]$. These predictions were borne out by Tazaki's (2026) study. Tazaki (2026) conducted an acceptability judgment task with Japanese L2 English learners. In his experiment, two types of verbs were included: non-alternating causative verbs that lexically specify $[+C]$ but not $[\pm V]$ and $[\pm S]$ and non-alternating transitive verbs, most of which lexically specify $[\pm V]$ and $[\pm S]$. These two types are referred to as causative and transitive verbs. The results of this study showed that intermediate learners were less accurate at rejecting non-alternating causative verbs in the non-target anticausative construction than at rejecting transitive verbs in the same construction.

The findings can be explained if non-alternating causative verbs are erroneously classified as alternating due to the input, as in (29), where non-alternating causative verbs are used with subjects that do not have to be $[+V, +S]$. As inputs like (29) indicate that the verbs *destroy* and *kill* follow the generalization (28), they are subsumed under the causative alternation verbs (see also Tazaki, 2026, for another explanation of the results).⁹

In addition to the overgeneralization errors with non-alternating causatives, Oshita (2002) observed L2 English learners' resistance to accepting grammatical anticausatives. In his acceptability judgment task, Italian and Japanese learners of L2 English rated sentences with alternating verbs (e.g., *melt*, *shatter*) in NP-V word order less accurately than those with unergatives, although both types of sentences are grammatical in English. Their poorer performance with alternating verbs can be interpreted as reflecting a misanalysis of the Proto-Role properties associated with these causative variants. In the L2 grammar assumed for the learners in Oshita's (2002) experiment, alternating causative verbs were erroneously analyzed as belonging to the non-alternating class and causative

⁹ In the literature, causativization errors have also been reported, as discussed in Section 3.2.1. I leave for future studies the question of how these errors are explained under the proposal.

verbs that specify the values of $[\pm V]$ and $[\pm S]$. If this analysis is correct, the Theta Misanalysis Hypothesis can account for learners' failure to accept grammatical anticausative sentences

This section has discussed errors concerning the causative alternation and suggested that the current proposal can be extended to the acquisition of the causative alternation. The Theta Misanalysis Hypothesis is not necessarily specific to non-target passivization and anticausativization. Rather, it may provide a potential explanation for a range of VP-level phenomena related to theta assignments.

7. Conclusion

This study begins by reviewing prior analyses of passive unaccusative errors and critically assessing their explanatory power (Section 3). The Conceptualizable Agent analysis and subject animacy analysis rely on the problematic assumption that passive unaccusative errors stem from non-target lexical causativization. While the overt marking of NP-movement analysis addresses some issues associated with this assumption, it fails to account for the previous observations, namely, the effects of discourse-pragmatic contexts and subject animacy (agentivity). To overcome these limitations, this study proposed a novel account—termed the Theta Misanalysis Hypothesis—which builds on the NP-movement Analysis (Section 4). This hypothesis posits that theta roles are not discrete primitives but cluster concepts, following Dowty (1991), and that Proto-Role properties in L2 grammars are shaped by a discourse-pragmatic context and subject semantics.

Section 4.2 extended this proposal to the phenomenon of passive unergatives. An acceptability judgment task was conducted to test the predictions of the Theta Misanalysis Hypothesis empirically. The results revealed significant effects of lexical class but no significant effects of discourse-pragmatic context. Specifically, passive constructions with change-of-state/location verbs (unaccusatives) were rated as more acceptable than those with non-controlled process verbs (unergatives), which, in turn, received lower ratings than those with controlled process verbs (also unergatives). This pattern indicates that discourse context does not influence passivization errors in L2 English; nonetheless, the Theta Misanalysis Hypothesis remains compatible with these findings (Section 5).

Most importantly, the results of this experiment cannot be accounted for by any of the three analyses—the Conceptualizable Agent, subject animacy, or overt marking of NP-movement—but only the Theme Misanalysis Hypothesis does.

In summary, the Theta Misanalysis Hypothesis offers a coherent and empirically grounded account of passivization errors with intransitive verbs in L2 English. By rejecting the assumption of lexical causativization and focusing on misanalyses of argument structure and Proto-Role properties, it captures a wide range of empirical phenomena, including passive unaccusatives and unergatives. Moreover, as it does not rely on L1 transfer, this hypothesis yields universal predictions that may extend to the acquisition of unaccusativity in other L2 languages.

Acknowledgements

I would like to thank Hiroyuki Shimada, Tatsuya Suzuki, Hiroyuki Oshita, Takayuki Kimura, Shigenori Wakabayashi, Makiko Hirakawa, Taiki Mochizuki, and Satoshi Hattori for their valuable comments and discussions. I am also grateful to the editor of *Second Language* and the three anonymous reviewers for their helpful comments and suggestions.

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Appendix

List of experimental sentences with contextual paragraphs (experimental sentences were italicized)

Change-of-state/location (unaccusative)

1. Kevin bought a car 30 years ago. He kept driving the same car for 29 years, but last year, the old car finally stopped working. (Internal Causation)
The engine died / was died last year.
2. This morning, David, who loves cars, was working on the engine of a new car with his child. While David looked away for a moment, the child hit the engine with a nearby hammer. As a result, the engine completely stopped working. (External Causation)
The engine died / was died in the morning.
3. Emily's ring slipped off her finger and rolled down the slope. Unfortunately, the ring sank into the huge lake. (Internal Causation)
The ring disappeared / was disappeared into the huge lake.
4. The magician tightly clenched the coin in his right hand. Then, he struck his right hand, which held the coin, with his left hand. The result amazed the audience. (External Causation)
The coin disappeared / was disappeared from his right hand.
5. Yesterday afternoon, it suddenly started raining but stopped shortly after. When the rain stopped, a beautiful rainbow became visible in the sky. (Internal Causation)
The beautiful rainbow appeared / was appeared in the sky.
6. The magician pointed to an empty table. The next moment, he struck the table three times with a stick he held in his right hand. The result amazed the audience. (External Causation)

The coin appeared / was appeared on the table.

7. Jesse bought a discounted apple. After returning home, they placed it on the living room table. The table was slightly tilted, and without Jesse noticing, the apple rolled away. (Internal Causation)

The apple fell / was fallen from the table.

8. Meg's friend placed an expensive cup they had recently bought on the desk and started bragging about it. Feeling jealous, Meg pushed the cup to the edge of the desk, intending to break it. (External Causation)

The expensive cap fell / was fallen from the table.

Non-controlled process (unergative)

1. Sarah had been practicing hard to win the basketball tournament. However, she couldn't win yesterday's tournament, and she was so frustrated that she couldn't sleep last night. Her pillow was soaked with tears. (Internal Causation)

Sarah cried / was cried last night.

2. Two days ago, Ken and Naomi were having an argument. Ken started to get really irritated, and in the end, he hit Naomi. The pain brought tears to Naomi's eyes. (External Causation)

Naomi cried / was cried two days ago.

3. Two days ago, Kevin watched a very funny TV show. Yesterday, while he was at school, he suddenly remembered one of the funniest scenes, and it made him react uncontrollably. (Internal Causation)

Kevin laughed / was laughed at school.

4. Jack is very ticklish and reacts immediately even to the slightest touch. Three days ago, Jack was in a bad mood and ignored everything his brother did to him. However, when his brother tickled him, he couldn't help but react. (External Causation)

Jack laughed / was laughed three days ago.

5. While having lunch at a Mexican restaurant, Alice absentmindedly put her hand into the pocket of the pants she hadn't worn in a long time. Inside the pocket, she found a 10,000-yen bill she thought she had lost. Alice was overjoyed. (Internal Causation)

Alice smiled / was smiled in the Mexican restaurant.

6. Brian was in a bad mood all day, and his family didn't know what to do about it. To cheer him up, his close brother went to Brian, who was sitting alone in a quiet room, and told him one of his favorite dad jokes. (External Causation)

Brian smiled / was smiled in the quiet room.

7. Leonardo participated in a marathon and spent the entire day running under the summer sun. Completely exhausted, he collapsed onto his bed as soon as he got home, his legs giving out. When he woke up, it was already morning. (Internal Causation)

Leonard slept / was slept on the bed.

8. A small tumor was discovered in 4-year-old Bruce's body, and it was decided that surgery would be performed to remove it. However, Bruce resisted and ran wild in the hospital, so the doctor administered an anesthetic to calm him down. (External Causation)

Bruce slept / was slept in the hospital.

Controlled process (unergative)

1. Mike exercises outside every morning. However, since it was raining yesterday, he couldn't go outside. Instead, he used the treadmill at home to get his exercise. (Internal Causation)

Mike walked / was walked on the treadmill.

2. Bob was told by his wife to lose weight, and yesterday she took him to the gym. He was forced onto a treadmill and was watched by his wife the entire time to make sure he didn't slack off. (External Causation)

Bob walked / was walked on the treadmill.

3. Mary participated in a dance contest being held today. She strongly wanted to win the contest and practiced a lot until the day of the competition. (Internal Causation)

Mary danced / was danced in the contest.

4. Mary hates physical activity. Yesterday, one of her friends suggested filming a TikTok together and grabbed her arm, forcing her to move. (External Causation)

Mary danced / was danced yesterday.

5. John goes to a karaoke box with his friends every Sunday. Last Sunday, his friends couldn't come because of work, so he went alone. (Internal Causation)
John sang / was sung last Sunday.
6. Jessica hates going to karaoke boxes and would just sit quietly even when her friends took her there. However, when her friends took her last week, they forced her to hold the microphone, and she couldn't refuse. (External Causation)
Jessica sang / was sung last week.
7. Six-year-old Bill, who recently learned to read, has started taking an interest in science. Yesterday, he went to the library and read many science books. (Internal Causation)
Bill studied / was studied at the library.
8. Tony, a high school student, got a terrible score on last month's math test. When Tony's mother found out about the result yesterday, she was furious and refused to give him dinner until he finished his math workbook. (External Causation)
Tony studied / was studied yesterday.